

Impacts of birthrate decline



Overview

- In 2023, the total fertility rate in England and Wales was 1.44 children per woman, the lowest on record. Because of declining fertility rates, and other factors such as people living longer, the UK has an ageing population.
- Population changes can affect the number of people requiring services, and the number of people available to provide services, including in health and social care. They can also affect the number of younger and older people in the workforce.
- Workforce changes may affect the economy, businesses and individual workers. Wider potential implications of birthrate decline may include effects on living standards, the provision of informal care, and the environment.
- Some commentators suggest immigration can contribute to short-term increases in the total fertility rate. Some contend migration alone is not likely to play a significant role in alleviating pressures of an ageing population.
- The global fertility rate has been declining almost continuously over the past 50 years. Some countries have implemented policies to improve family/work balance, which may support having and raising children. Evidence on the effectiveness of policies which seek to raise fertility is limited and contested.

Background

To keep the same population size without positive net migration, the total fertility rate (TFR) for countries with low mortality needs to be around 2.1 children per woman.¹ The rate has been lower than this in England and Wales since the 1970s.¹ Reporting on 2023 data, the Office for National Statistics (ONS) noted that the TFR in England and Wales had fallen to 1.44 children per woman, the lowest levels on record.^{1,a}

The population of the UK is ageing. The ONS says this is primarily driven by people living longer,^b having fewer children and having children later in life.⁴ Another contributing factor is that people born in periods of high fertility are now reaching older age groups.⁵ In 2025, the House of Lords Economic Affairs Committee launched an inquiry into 'Preparing for an Ageing Society'.⁶

The ONS produces population estimates, and projections of the future size and age structure of the population of the UK. Population projections are based on current trends. These are not forecasts and will differ from the actual future population.⁷

By mid-2047, the ONS projects the number of people aged 85 years and over to be nearly double the number in mid-2022.⁸

Over the 10 years between mid-2022 and mid-2032, the population of the UK is projected to increase by 4.9 million (7.3%).⁸ This increase is projected to arise from net migration, and it is projected that there will be a similar number of deaths and births in the UK.⁹ As noted in a 2024 House of Commons Library article, the UK's growing and ageing population may impact public services, such as in healthcare and education.¹⁰ Population change may affect the number of people that require services and the number of people available to provide them.¹⁰

Some academics have expressed concern that a declining birthrate (see Box 1), in the context of an ageing population, will result in a smaller workforce available to fund public services.¹¹

In 2024, the Department for Work and Pensions reported that the average age at which people stop working is increasing.¹² Some academics have suggested that extending working lives may mitigate negative effects of the declining fertility rates and ageing population.¹³

The charity Population Matters^c says that there may be environmental benefits to population reduction through global birthrate decline, but it notes the importance of other factors, such as the use of fewer resources.¹⁵

^a The ONS reported that in 2023, Wales and the North West of England saw the largest decreases in the TFR, whereas London, the North East and the West Midlands saw the smallest decreases.²

^b Commentators note that this is due to improvements in nutrition, hygiene, housing, sanitation, and public health measures.³

^c Population Matters is a charity campaigning to achieve a sustainable human population. It sees "population stabilisation as a necessary condition for a sustainable future in which *all* of humanity thrives".¹⁴

This POSTnote evaluates current demographic trends, estimates and projections in England and Wales. There is uncertainty in population projections estimates, and financial forecasts^d, which means that actual outcomes may differ and estimates should be considered in light of this uncertainty.^{8,17} Research in this area may also draw on projections and forecasts about the future.

This POSTnote considers the potential impacts of demographic change on the workforce and public services, as well as wider social, economic and environmental impacts. It covers contested topics that often lack consensus.

This briefing focuses primarily on the demographic context in England and Wales, unless otherwise stated. However, some information may be relevant to the UK more broadly. This POSTnote does not cover the potential causes of declining fertility levels, or policy responses that may influence the birthrate in the UK. It does cover some wider international context, including on responses to declining fertility levels.

Current and projected demographic trends

Declining total fertility rate

In 2023, England and Wales recorded the lowest TFR (see Box 1) since records began in 1938.² The TFR was 1.44 children per woman, lower than the 2022 TFR of 1.49.¹ This is slightly below the 2022 Organisation for Economic Co-operation and Development (OECD)^e average of 1.5.¹⁹

^d Forecasts, such as those provided by Office for Budget Responsibility, attempt to predict the impact of political and economic changes.¹⁶

^e Organisation for Economic Co-operation and Development (OECD) “is an international organisation that works to build better policies for better lives”.¹⁸

Box 1: Total fertility rate (TFR)^f and birthrate

Total fertility rate (TFR)

The TFR is a measure of the average number of children a woman would have in their lifetime based on current birth rates.²¹

The TFR is a snapshot of fertility rates in one specific year.²¹

Birthrate

Birthrate refers to the number of live births per thousand of population per year. It measures the fertility of a given population as a whole.²²

This POSTnote draws on sources which refer to both declining total fertility rates, and declining birthrates. These demographic trends are often, but not always, seen alongside each other.

ONS data show that, in 2023, in England and Wales, there were 591,072 live births, a decrease of 14,407 live births compared with the previous year (605,479).² The ONS state that this is the lowest number of live births on record since 1977 (569,259).²

The ONS also say that, since 2012, there has been an overall decline in births in England and Wales. ONS data shows that, whilst in 2021 there was a brief increase in live births⁹, the number of live births fell again in the years 2022 and 2023.²

Figure 1 shows how the number of live births has changed from 1900 to 2023.

ONS data shows that, in England and Wales, women are starting to have children later in life than previous generations.²³ There has also been a decrease in younger women having children; between 1964 and 2023, there was a 79% decrease in live births per 1,000 women aged 20 to 24 years.²

In 2025, the ONS projected that women in England and Wales will continue to have children later in life and have fewer children than previous generations.²³

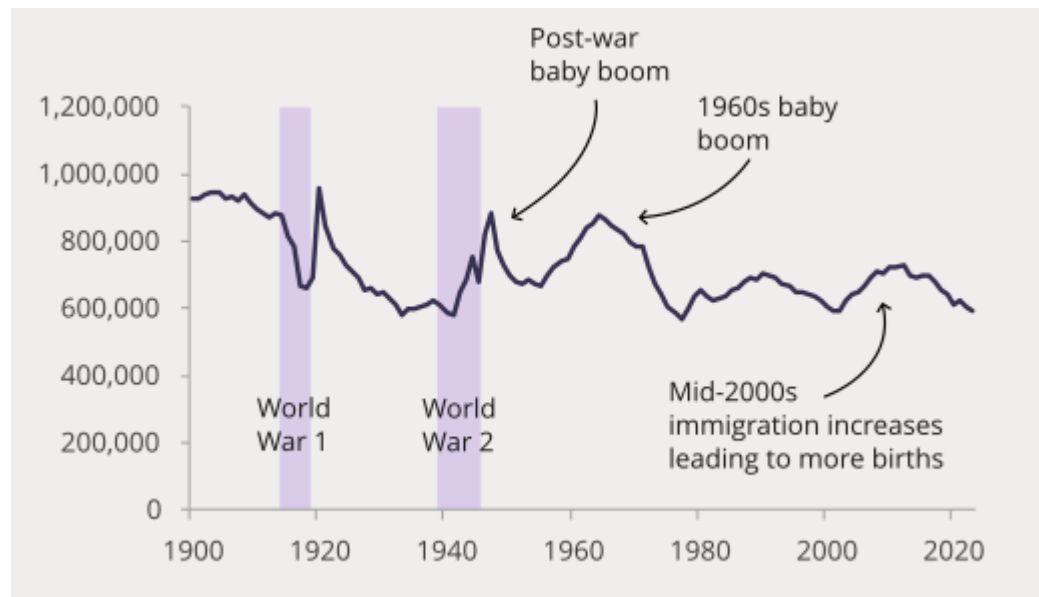
Experts have noted that the “completed family size”, which is how many children a woman has had by age 45, has remained relatively stable. They note it is unclear for the current population under 45 if they will have children later, meaning they will

^f The official definition from the World Health Organisation of TFR is: “The average number of children a hypothetical cohort of women would have at the end of their reproductive period if they were subject during their whole lives to the fertility rates of a given period and if they were not subject to mortality.”²⁰

⁹ The ONS state this brief increase was likely because of the effect of the covid-19 pandemic.²

have a similar completed family size, or have fewer children, or no children, meaning a reduction in completed family size.²⁴

Figure 1 Number of live births, England and Wales, 1900 to 2023



Source: [Births in England and Wales from the Office for National Statistics](#)^{2, h}

Evidence suggests that causes of birthrate decline and declining fertility are complex. Stakeholders present a variety of social,^{25,26} personal,^{27,28} and economic factors^{19,29,30} that may contribute to these trends.^{ij}

Ageing population

The UK population is ageing. Between mid-2022 and mid-2032, the ONS projects (based on 2022 data):

- the number of children^k in the UK to decrease by 797,000 (negative 6.4%), from 12.4 million to 11.6 million
- the number of people of pensionable age in the UK to increase by 1.7 million (positive 13.8%), from 12.0 million to 13.7 million⁸

Some academics have contended that population ageing is driven more by falling fertility rates than by falling mortality rates.³⁸

^h Information formatted by Cassie Barton, House of Commons Library, 1/4/2025

ⁱ The ONS reported that the covid-19 pandemic affected mortality rates.³¹ Some evidence suggested a 14.1% decline in live births in January 2021 compared with January 2018 and 2019.³² Some academics suggest the decline in birth rates in the later phase of the covid-19 pandemic was associated with economic uncertainty.³³

^j Some people may choose not have children because they do not want children.^{34,35} Some may want (more) children but choose not to have them.^{36,37}

^k Children are defined as those aged 0 to 15 years.

Using mortality data from the 2022-based national population projections, the ONS projects life expectancy for children born in 2047 will increase, as displayed in table 1.³⁹

Table 1 Projected cohort life expectancy^l in the UK (years)^{39,m}

	Born in 2023	Born in 2047	Difference
Girls	90.0	92.2	2.2
Boys	86.7	89.3	2.6

A 2024 policy briefing by the Centre for Population Changeⁿ suggested that rural and coastal areas of the UK are experiencing “the most rapid ageing”.^{40,o}

The World Health Organisation has noted advantages of an ageing population, including opportunities those in later life can take up, and the contributions older people make to their families and communities.⁴²

[Healthy ageing and care for older populations](#) is discussed further in POST’s 2024 horizon scan article.

Ethnic composition

The ethnic composition of England and Wales is changing. Based on 2021 census data, the ONS reported that “the percentage of the population in all high-level ethnic groups, excluding “White”, has increased since 2011”.^{43,p}

^l Cohort life expectancy is the average number of years people born in each year are expected to live, taking into account assumed future changes in mortality over their lifetimes. (Period life expectancy ignores future changes in mortality rates: it is the average number of years a person would live if they kept the same age-specific mortality rates of a given time period and area for the rest of their life.)³⁹

^m Data from [Past and projected period and cohort life tables - Office for National Statistics](#) (2022-based, UK, 1981 to 2072)

ⁿ The Centre for Population Change is a research centre and includes Universities of Southampton, St Andrews, and Stirling. The organisation investigates population change and is primarily financed by the Economic and Social Research Council.

^o A 2021 Centre for Ageing Better report suggested that this is often driven by older people moving into rural areas as younger people move out of them.⁴¹

^p The ONS says “many factors... may be contributing to the changing ethnic composition of England and Wales, such as differing patterns of ageing, fertility, mortality, and migration”.⁴³

Migration and birthrate decline

The Office for Budget Responsibility's (OBR's)^q 2025 forecast estimates that net migration into the UK will decrease from 728,000 in mid-2024 to 258,000 in mid-2027.^r It is then forecast to rise to 340,000 by mid-2030.⁴⁴

A 2016 journal editorial suggested that immigration into the UK and Europe would contribute to:

- short-term increases in the TFR
- population growth
- growth in the workforce⁴⁵

Commentators have suggested that immigration can lower the "old-age dependency ratio", as migrants are often of working age, and tend to have more children than UK nationals.^{s,46} ONS data shows that in 2023, 31.8% of all live births were to non-UK-born mothers (187,975 births were to mothers were born outside the UK, out of a total of 591,072 births in total) in England and Wales (an increase from 30.3% in 2022).⁴⁷

However, in 2001 and 2014 some academics suggested immigration may be an unsustainable solution to population decline and ageing in the long term, as immigrants begin to have fewer children and age themselves.^{48,49,t, u}

Selective migration policies aim to allow immigration when specific criteria are met, such as age or skill set. These policies often favour highly skilled workers with the aim of filling gaps in the labour market.^{51,52}

The UK Government has processes to encourage the migration to the UK of people with certain skillsets.^{v,53–55} Considering ageing populations across many countries, some academics have concluded that, although large-scale, selective immigration

^q The OBR is an independent body that examines UK public finances.

^r The OBR suggests that decreases in net migration over the forecast are "consistent with the tightening of visa policies... and higher levels of emigration following recent high levels of immigration".⁴⁴

^s The Organisation for Economic Co-operation and Development (OECD)^s defines "old-age dependency ratio" as the number of people aged 65 or over for every 100 people of working age (20 to 64 years old).

^t OECD reports from 2024 and 2023 say that the "net contribution of migrants to overall fertility levels is relatively small" and that "fertility patterns among migrants and the native-born tend to converge over time and across generations".^{19,50}

^u For more context, please see: [Economic Affairs Committee - Corrected oral evidence: Preparing for an ageing society](#)

^v For example, the Global Talent visa aims to encourage "exceptionally talented people" in areas of academia, arts and culture and digital technology to live and work in the UK.^{53–55} In 2025, the Government's white paper 'Restoring Control over the Immigration System' said it would make it "easier for top scientific and design talent to use our Global Talent visa".⁵⁶

may alleviate pressures arising from an ageing population, migration in isolation may not play a significant role in doing so.⁵⁷

The ONS reported that higher levels of net migration will slow population ageing in the UK but will not prevent it.^{58,w}

In 2025, the Government released the 'Restoring Control over the Immigration System' white paper. The Government said it outlined policies "reversing the long-term trend of increasing international recruitment at the expense of skills and training".⁵⁶

Potential impacts of declining fertility rates

Changes to the workforce

Birthrate decline and an ageing population may mean fewer younger, and more older, workers.⁶⁰ Because of the increase in life expectancy and increases to State Pension age, some evidence indicates people may remain in the workforce for longer, by choice or necessity.^{60,61,62} ONS data shows that between April to June 2022, employment of those aged 65 or over in the UK reached record highs.⁶³

The ONS projects that the old-age dependency ratio is to increase over the next 40 years.⁵⁸ The [ONS explorer tool](#) calculates this for people aged 65 and over, but the projections data explores how this changes as people work beyond 65 years.⁶⁴

Some academics have observed a common misconception that older workers are less productive, leading to concerns for productivity because of the ageing population.⁶⁵ A 2014 report by Age UK^x said that evidence suggests either a lack of a relationship between productivity and age or that older workers are at least as productive as their younger colleagues.⁶⁷

A 2022 Chartered Institute of Personnel and Development (CIPD)^y report noted that, because of technological advances in the workplace, workers of all ages will need to upskill or reskill during their career.^{69,z}

^w In 2019, the Head of the ONS Centre for Ageing and Demography noted in relation to the old-age dependency ratio that "changes in SPA [State Pension age] and increases in the numbers of older workers have more of an impact on population ageing compared with international migration".^{46,59}

^x Age UK is a charity aiming to provide support for older people and drive change for older people to feel included and valued.⁵⁸

^y The Chartered Institute of Personnel and Development is a professional body for human resources, learning and development, organisation development and all people professionals, calling for good work that creates value for everyone.⁶⁸

^z The CIPD report also suggested that employer and public investment in workplace training and lifelong learning had been declining.⁶⁹

In 2024, research by the British Standards Institution^{aa} found that age-diverse workplaces provide opportunities to boost businesses and economies.^{71,bb} A 2020 OECD report suggested that more experienced workers can improve team performance through their contribution and interaction with others.⁷⁴

The Department of for Work and Pensions reported in 2024 that the most common reasons why people aged 50 to 64 years were not active in the workforce were that they were sick, injured or disabled.^{12,cc} In 2025, the Centre for Ageing Better said that “nearly half a million people aged 50 to 65 who have left work due to health reasons would return if the right support were in place”.^{77,dd}

Workforce changes and employees

The House of Lords Select Committee on Public Service and Demographic Change 2013 report ‘Ready for Ageing?’ suggested that, in addition to economic benefits, working for longer may have individual health, social and intellectual benefits.⁷⁹ Some academics have argued that improvements in work quality and security should be a condition for supporting policies for longer working lives.⁸⁰

Commentators have noted that some people may wish to retire earlier but may not be able to afford to do so.⁸¹ Academics have suggested many factors associated with preferred retirement age, including self-reported health, work ability, job satisfaction and income.^{82–84}

A 2025 Centre for Ageing Better report said that although health is often a reason older people leave the workforce, age discrimination “has a huge impact on someone’s chances of being in employment”.^{77,85} A 2021 report from the same organisation suggested that ageism existed at each stage of job recruitment and harmed older workers’ job prospects.⁸⁶

Workforce changes and businesses

Some organisations have indicated that businesses need to adapt to the changing workforce.^{87,88} This may include supporting the employment of older workers.⁸⁹ In

^{aa} The British Standards Institution is the UK's National Standards Body representing UK economic and social interests across all European and international standards organisations.

^{bb} Some commentators note the risks of age discrimination in the workplace which can arise when multiple generations work alongside each other.^{72,73}

^{cc} Some researchers have noted that some people, particularly women, may undertake unpaid care work in their families or communities instead of undertaking paid employment in the workforce.^{75,76}

^{dd} A 2015 House of Commons Library briefing noted that challenges of the ageing population could potentially be mitigated if individuals remained healthier, and therefore in the workforce, for longer.⁷⁸

2015 the CIPD suggested supporting flexible working and employee health and well-being may help people maintain employment into older age.^{88,ee,ff,gg}

Other adaptations include strategies to fill the labour supply gap.^{88,93} Some academics contend that businesses could increase their workforce by providing childcare for employees.^{94,hh,ii}

Some academics say that immigration can play a role in increasing the number of working-age people.⁹⁷ The use of immigration to increase the supply of labour and skills is a contested topic.⁹⁸ Some academics have noted this can remove skilled professionals from other countries.⁹⁹ Others raise concerns around employers exploiting migrant workers.¹⁰⁰

A 2023 NHS Confederation article noted that immigration and international recruitment could help to fill the over 100,000 vacancies at that time in the NHS.¹⁰¹

In 2022 the Migration Advisory Committee^{jj} reported that “most employers”^{kk}, out of 28 interviewed, did not specifically use migration to manage skill and staff shortages.^{102,ll,mm}

^{ee} A 2019 report by the British Medical Association said that flexible working was the most important way to support older doctors.⁹⁰

^{ff} A 2024 article for the Safety Hub, a blog about health and safety emphasised the role of ergonomic adjustments in the workplace in meeting the needs of an ageing workforce.⁹¹

^{gg} The CIPD also noted that many people with disabilities or long-term health conditions would like to work and can offer their skills and talents but lack the support necessary to enter or remain in the workforce.⁹²

^{hh} Others suggest that declining fertility rates increase the number of women in the workforce which can contribute to economic growth.⁹⁵

ⁱⁱ The World Bank has suggested that closing the gender gap in employment and entrepreneurship could increase global GDP by over 20%.⁹⁶

^{jj} The Migration Advisory Committee is an independent public body. It advises the government on migration policy.

^{kk} Employers were from three sectors: construction (10), manufacturing (10) and IT (8).¹⁰²

^{ll} However, this did occur in some circumstances where this did occur such as Intra-Company Transfers or bulk recruitment.¹⁰²

^{mm} Employers reported that where migrants did join their workforce, this was through approaches such as: ‘formal/traditional recruitment processes’, ‘within-company transfers or rotations’ and ‘recruitment through networks’.

Role of technology

Potential reductions in productivity from a smaller and/or ageing workforce may be averted by technological advances, for example, the increased use of artificial intelligence (AI), automation or robotics.^{93,nn} A 2025 UN report noted that AI may increase productivity and income for some workers, but may lead to job losses for others.^{104,oo}

In 2025, the Institute for the Future of Work^{pp} reported that almost 80% of 1,000 UK firms it surveyed had adopted these technologies.¹⁰⁶

Research in 2025^{qq} found that the impact of technology adoption on workers is mixed. Many workers report feelings of anxiety and powerlessness in response to technological change. The research also suggested that advances have increased efficiency, flexibility and collaboration, but the impact on work intensity, work-life balance and job security varies.¹⁰⁶

Workforce changes and the economy

Some stakeholders suggest that birthrate decline may result in a smaller workforce contributing to the economy, and therefore less funding for public services.^{rr} They say it may reduce economic growth in terms of gross domestic product (GDP) per person,^{ss} while demand in healthcare increases because of the higher number of older people.^{11,93,110,tt}

The average age at which people stop working is increasing. In 2022 the ONS reported the reasons were due to an increase in part-time work and self-employment in people aged 65 years and over.⁶³ In 2017, Age UK estimated that those over 65 years made economic contributions to society including:

- £54 billion from employment contributions

ⁿⁿ A 2024 news article suggested that AI could help prevent a decline in growth for the US economy as a result of birthrate decline.¹⁰³

^{oo} The 2025 United Nations report also said that it may take "years or even decades for the full extent of AI-related impacts to materialize".¹⁰⁴

^{pp} The Institute for the Future of Work is an independent research and development Institute, evaluating how technologies are transforming work and working lives.¹⁰⁵

^{qq} This research was for the Pissarides Review, which is a three-year programme of work involving collaboration between the Institute for the Future of Work, Imperial College London and Warwick Business School.¹⁰⁷

^{rr} This is based on the current tax system where "overall tax rates levied on employment income are higher... than the overall (personal and corporate) tax rates levied on income from running a business".¹⁰⁸

^{ss} Gross domestic product (GDP) per capita is an economic metric that breaks down a country's economic output to a per-person allocation.¹⁰⁹

^{tt} Some academics suggest that an increase in birthrates also reduces the workforce because of parental leave and increases in part-time employment. A report by the Government Equalities Office in 2019 found that one in five mothers return to and remain in full-time work in the first three years after having a child.¹¹¹

- informal childcare worth £7.7 billion^{112,uu}

Pensions

The ONS projects that between mid-2022 and mid-2032, the number of people at State Pension age will increase by 13.8%.⁸ According to the OBR, the State Pension is the largest single component of welfare spending.¹¹⁴ HM Treasury statistics from 2024 show that pension spending in the UK increased to approximately £140.6 billion in 2023/24, from £123.7 billion in 2022/23, due to both the ageing population and the triple-lock policy.^{115,116,vv}

The OBR has projected that from 2028/29 to 2073/74, spending on the State Pension is expected to increase by 2.7% of GDP.^{116,ww}

Income tax

There is limited evidence around how the birthrate decline may affect income tax.

In 2025, the think tank Policy Exchange modelled that the average income tax paid by people in Great Britain might need to increase between 1.4 to 7.4 percentage points^{xx} per taxpayer by 2080 to stabilise the budget because of birthrate decline and the ageing population.^{121,yy}

Impacts on provision of public services

The declining birthrate may impact demand for and provision of public services, including education and healthcare.

Education

The ONS projects that by mid-2032 there will be fewer children in the UK.⁸ Between mid-2022 and mid-2032, the ONS projects (based on 2022 data) the number of children^{zz} in the UK to decrease from 12.4 million to 11.6 million.

^{uu} Research by the charity Age UK in 2017 found that around 40% of grandparents in the UK provide regular childcare for their grandchildren.¹¹³

^{vv} The [triple lock policy](#) is a government commitment to uprate the basic and new State Pension annually by the highest of earnings, prices or 2.5%.¹¹⁷

^{ww} Regarding private pensions, some stakeholders have noted that an increasing preference in recent years for defined contribution pensions, rather than defined benefit pensions, moves the risk of a shortfall in money in pension pots from employers to individuals. There are two main types of private pension: defined contribution, based on how much is paid in, and defined benefit, based on your salary and how long you've worked for an employer. ¹¹⁸⁻¹²⁰

^{xx} This estimated percentage point increase varied based on migration levels and birthrate:

- High-Immigration, Low-Birth: 4.1
- Low-Immigration, Low-Birth 7.4
- Mid-Immigration, High-Birth 1.4¹²¹

^{zz} Children are defined as those aged 0 to 15 years.

In 2024 the Department for Education projected that in England in 2028, there would be 207,000 fewer nursery and primary school pupils than in 2024.¹²² The amount of funding that schools in the public sector receive from the government is partly determined by the number of pupils attending the school.¹²³

A 2024 House of Commons Library article noted that the declining birthrate could be one of several factors reducing the demand for primary school places in London.¹²⁴ Other factors include changing immigration patterns, an increase in home education, and financial pressures resulting in people moving to more affordable areas of the UK.^{124,aaa}

A 2024 report by London Councils^{bbb} estimated that the demand for reception^{ccc} places in London would fall by 4.4% between 2023/24 and 2027/28 and that demand for year 7^{ddd} places could fall by 10.9% in some boroughs.¹²⁹ The report said that having fewer children in the population is creating additional challenges for maintained primary schools in London, a quarter of which are in financial deficit.^{129,eee}

Some schools have had to merge or close due to falling pupil numbers.^{129,131} Some parliamentarians have said that, as a part of school management, spare capacity may need to be reduced or repurposed, and have noted the role of local authorities in managing “periods of decline, bulges and shifts in local patterns”.¹³²

The OBR has projected that from 2028/29 to 2073/74, spending on education will decrease by 0.5% of GDP, partly driven by the declining birthrate.^{116,fff}

Some commentators see this as an opportunity to improve education quality. They suggest investing in schools to offer smaller classes and more targeted support rather than reducing funding.^{134,135}

Some academics have suggested that the reduced pupil numbers may offset the increased cost of the ageing population.¹³⁶

Health and social care

In 2024, Lord Darzi’s report, an independent investigation of the NHS in England, noted the ageing population was the most significant driver of increased healthcare

^{aaa} According to the Department for Education, the demand for school places varies depending on the area in UK; for example, primary pupil numbers are expected to increase in the East Midlands.^{125,126}

^{bbb} London Councils is a collective of the London local government (32 boroughs and the City of London Corporation).¹²⁷

^{ccc} Reception is the first school year in England and Wales. Most children “start reception full-time in September after their fourth birthday.”⁶⁶

^{ddd} In England and Wales, children aged 11 to 12 are in year 7, which is the first year of secondary education.¹²⁸

^{eee} The report also said that stretched budgets may lower education standards, and affect attainment and support for pupils with special educational needs and disabilities.^{129,130}

^{fff} A 2024 report by the Education Policy Institute, a research institute, suggested that by 2029/30 schools in England could lose up to £1 billion in funding.¹³³

needs due to long-term health conditions.^{137,999} The 2024 OBR long-term projections, which consider public finances over the next 50 years^{hhh} suggest that the ageing population, life expectancy, and healthy life expectancy will drive 0.6 percentage points of the average annual real growth (3.1%) in health spending.¹³⁹

Between 2017 to 2019 (the period before the covid-19 pandemic) and 2021 to 2023, healthy life expectancy at birth in England¹⁴⁰ and Wales fell.¹⁴¹

The 2013 'Ready for Ageing?' report said the NHS needed to adapt to cope with increased demand and costs.^{79,iii}

In 2023, Skills for Care^{jjj} estimated that the adult social care system in England will need 25% more paid care workers by 2035,¹⁴³ which would more than double the current cost of paid care services.^{143,kkk,III}

In its 2025 immigration white paper, the Government said it would "close social care visas to new applications from abroad." It added that, until 2028, ... they would "permit visa extensions and in-country switching for those already here", although this would be kept under review."⁵⁶

Some academics contend that it is proximity to death, rather than age, that determines health care expenditure in older populations.^{147,148} Others have contended that healthcare sector costs depend on how morbidity^{mmm} changes over time.¹⁵⁰

A 2024 report by Age UK said that it was "too simplistic" to conclude that an ageing population means a "greater burden of disease and disability".¹⁵¹

A 2021 NHS Englandⁿⁿⁿ report stated that "over the past decade, workforce growth has not kept up with need, and the way staff have been supported to work has not kept up with the changing requirements of patients".¹⁵³

A 2023 report by ReWAGE^{ooo}, suggested that the NHS in England has become increasingly dependent on overseas health professionals.¹⁵⁴ According to a 2023

⁹⁹⁹ In 2021 the Health Foundation think tank reported that long-term conditions increase with age and, by 85 years, only 14% of people report having no long-term condition.¹³⁸

^{hhh} The OBR state these long-term projections are "highly uncertain".¹³⁹

ⁱⁱⁱ A 2021 House of Commons Library briefing noted that people's housing needs change as they age, and that literature had identified a shortage of accessible and specialist housing for older people.¹⁴²

^{jjj} Skills for Care is a strategic workforce body for adult social care.

^{kkk} In 2024, the National Centre for Social Research, (an independent not-for-profit social research organisation.¹⁴⁴) reported that their data indicates that Gen Z (born between late 1990s and the early 2010s) is the only generation where a majority want to see the government cover all adult social care costs.¹⁴⁵

^{III} In 2024, the King's Fund estimated that around half of the total adult health care expenditure in England was on people aged 65 years or older.¹⁴⁶

^{mmm} Morbidity is defined as the prevalence of a disease/illness/medical condition.¹⁴⁹

ⁿⁿⁿ It was announced in March 2025 that NHS England will be abolished and brought back into the Department of Health and Social Care.¹⁵²

^{ooo} ReWAGE is an independent advisory group to the government.

House of Commons Library briefing, 18.7% of the NHS workforce reported a nationality other than British.^{155,ppp} The Health Foundation think tank said in 2019 that "a non-restrictive migration policy will be crucial in ensuring the numbers of doctors, nurses and social care workers are sufficient to provide high quality care".¹⁵⁶

A 2025 press release reported that a maternity unit in north London is due to close due to a 14% reduction in the number of births over the past five years.¹⁵⁷

Wider social implications

Living standards

A 2024 article by the Economics Observatory^{qqq} suggested that the UK's annual productivity growth may need to increase to 1.5%, from the current rate of 0.5%, to strengthen improvements in living standards to above what they have been in the past decade.¹⁵⁹

Some academics have suggested that a high population growth will lead to a decline in the average standard of living, and risks a "continuously increasing unevenness of access to resources".^{160,rrr}

Some commentators contend that a smaller population would be beneficial because of the reduced pressure on resources, including housing, infrastructure and services.¹¹⁷

Others contend that fewer people could limit economic growth and attainment of higher living standards.¹⁶²

Provision of informal social care

Social care is often provided informally by unpaid family members and friends.¹⁶³ In 2019, Age UK said that although the number of people providing informal care had increased between 2011 and 2015,^{sss} this increase had not met growing demand.^{163,164}

The report also stated that "smaller and more dispersed families, longer working lives, women's rising participation in the labour market and a growing number of people ageing without children means that it is unlikely the supply of unpaid care will increase to meet that need".^{ttt}

^{ppp} 35% of doctors and 27.2% of nurses reported a nationality other than British.¹⁵⁵

^{qqq} The Economics Observatory states that it "bridges the gap between academic research, government policy and the general public" aiming to "provide balanced and reliable answers to questions about the economy".¹⁵⁸

^{rrr} Some academics have suggested that very low fertility undermines living standards, while moderately low fertility and population decline favour the "broader material standard of living".¹⁶¹

^{sss} The 2019 Age UK report suggested reasons for the increased pressure on informal care are the growing and ageing population, increasing complexity of needs and a decline in the numbers of people receiving social care services.¹⁶³

^{ttt} The 2019 Age UK report said "it is unrealistic to expect informal care to continue expanding indefinitely".¹⁶³

Wider economic implications

Some experts contend that a smaller workforce may result in a smaller number of entrepreneurs and innovators, which could be bad for the economy.¹⁶⁵

Others suggest the ageing population may create new opportunities. For example, the British Standards Institution has said that the increasing demand for healthcare has created “a surge in innovation across the healthcare system”.¹⁶⁶

A 2020 House of Commons Library briefing noted that technological advances in health care often result in spending increases.¹⁶⁷

Consumer market

Some academics have estimated that, by 2050, those aged 65 and over will account for one-quarter of global consumption, double that in 1997.⁹³ In 2019, the International Longevity Centre, estimated that by 2040, 63p of every pound spent in the UK will be spent by someone aged 50 years or over.¹⁶⁸

Wider environmental implications

The impact of population change on the environment is complex, context-dependent and contested.¹⁶⁹ A 2023 UN article said that environmental impacts, such as pollution, depend on human numbers and human activities.^{170,uuu,vvv}

Some academics and charities see the declining birthrate as potentially beneficial for the environment, as global population decline could reduce consumption. Population Matters contend that the population size has a direct, though complex and non-linear, effect on variables that contribute to biodiversity loss and climate change.¹⁵

Some researchers say there may be benefits to smaller human populations, including opportunities to reach some of the Sustainable Development Goals (SDGs),^{173,www} additional resources to tackle climate change,¹⁷⁴ and opportunities to avoid species extinctions.¹⁷⁵

Greenpeace and some academics have suggested that reducing the consumption of the richest countries would be more effective than reducing global population growth in reducing biodiversity loss and climate change.^{176,177}

^{uuu} The 2023 United Nations article also says that sustainability involves mitigating the environmental damage human activity causes.¹⁷⁰ According to Greenpeace, environmental sustainability “is a way of using resources that could continue forever”.¹⁷¹

^{vvv} A 2016 report by the United Nations Environment Programme said that “globally, growth in per capita income and consumption has been the strongest driver of growth in material use, even more important than population growth in recent decades”. The report says that increased material use results in climate change and increased biodiversity loss and pollution.¹⁷²

^{www} SDGs including SDG6 (Clean Water and Sanitation), SDG14 (Life under Water) and SDG15 (Life on Land).¹⁷³

International context

In 2024, the UN reported that “the world’s population is expected to continue growing over the coming 50 or 60 years”.¹⁷⁸ The UN expects nine locations^{xxx} to account for over one fifth of the projected increase in the world’s population between 2024 and 2054.¹⁷⁸

Global average TFR decreased from 4.8 in 1970 to 2.2 in 2024.¹⁷⁹ Figure 2 shows the total fertility rate worldwide between 1950 and 2023.

Global fertility decline is happening in different rates in different countries:

- In China, the Republic of Korea, Singapore and Ukraine, the total fertility rate^{yyy} was below 1.0 in 2024
- In countries with higher fertility levels, including Nigeria, Afghanistan, Sudan and Yemen, total fertility rates are projected to fall, with no country projected to have a total fertility rate greater than 4.0 in 2054.¹⁷⁹

Box 2 considers the impact of birthrate decline in the Republic of Korea.

A 2025 McKinsey Global Institute report said that, as a result of declining fertility rates, at different speeds and points in time the share of working-age people in the populations of all regions of the world^{zzz} will decline, which may have a number of economic implications, including on economic growth, consumption and saving patterns.⁹³

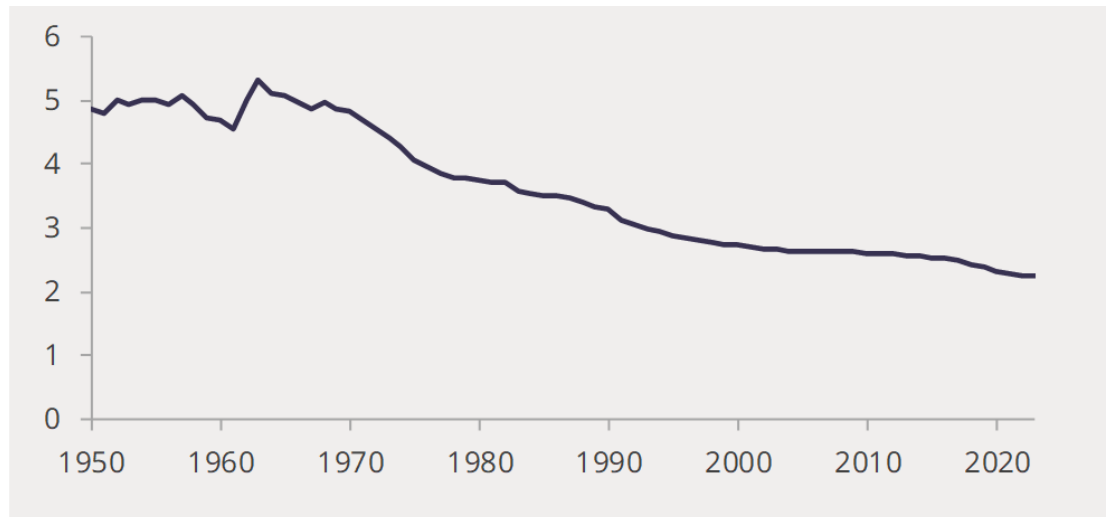
Some stakeholders have also said declines in working-age populations may have implications related to foreign policy including labour migration policy, international competitiveness, and productivity.^{50,93,180} For example, population changes could lead to countries competing to secure migrant workers.¹⁸¹ Others suggest potential changes to the “world order”, with some more populous countries “possibly emerging as new world powers”.^{181,182}

^{xxx} These nine locations are: Angola, the Central African Republic, Chad, the Democratic Republic of Congo, Mali, Mayotte, Niger, Somalia and the United Republic of Tanzania.¹⁷⁸

^{yyy} Information provided in the bullet points is from the United Nations World Fertility Report 2024 which states that “In this report, the terms “fertility rate”, “fertility level” and “total fertility” all refer to the total fertility rate (TFR)”.¹⁷⁹

^{zzz} The 2025 McKinsey Global Institute report consolidates the world into 10 regions to present findings.

Figure 2 Total fertility rate worldwide, 1950 to 2023 (births per woman)



Source: [United Nations World Population Prospects \(2024\)](#); [Human Fertility Database \(2024\)](#) – with major processing by [Our World in Data](#).^{134,aaaa}

Box 2: Case Study – Birthrate decline in the Republic of Korea

In 2023, the fertility rate in Korea was 0.72 children per woman, the lowest in the world.¹⁸³ In 2022, Statistics Korea^{bbbb} reported that the number of working-age individuals was projected to drop from 35.83 million in 2020 to 26.76 million in 2040.¹⁸⁴

In 2024, an OECD report suggested that the combined cost of health, long-term care and pensions is estimated to “more than double to 17.4% of GDP by 2060”.¹⁸⁵

According to a press release in 2024, 49 schools in Korea were scheduled to close because of the declining school population, a higher number than in previous years.¹⁸⁶

Some stakeholders reflect that that the projected population decline could mean natural resources could be used more sustainably.^{183,cccc}

Other stakeholders have raised the potential impact on the military, with the number of eligible men for conscription predicted to fall from 330,000 in 2020 to 180,000 per year by 2039.^{188,189}

^{aaaa} Information formatted by Cassie Barton, House of Commons Library, 25/3/2025.

^{bbbb} Statistics Korea is a government organisation that manages national statistics for Korea.

^{cccc} According to WWF-Korea, Korean residents have “an ecological footprint” eight times larger than their country’s ecosystems can provide”.¹⁸⁷

International responses

Some governments in countries with a low TFR^{dddd} have implemented policies to improve family/work balance, which may support having and raising children.^{190,191,eeee,ffff} Box 3 outlines some evidence regarding these policies between 2015 and 2019.

Evidence on the effectiveness of policies which seek to raise fertility is limited and contested.^{192,194,gggg}

Some academics and organisations suggest that government policies aimed at influencing birthrates can raise ethical questions, such as around who has agency and reproductive choice, and what people want for themselves in terms of family size.^{195,196} Many organisations emphasise the importance of gender equality, and that deciding whether to have, or not to have, children is a human right.^{14,197,hhhh}

Some academics have noted that policies to increase the birthrate may increase the number of dependants, with no workforce benefits for around 20 years.^{11,93} Others argue that international evidence suggests that policies to increase birthrates may affect the timing and spacing of births, not the total number of children born.^{11,198}

Box 3: International responses to declining fertility rates

A 2021 UN report showed that, that between 2015 and 2019:

- 55 governments had policies that “sought to raise fertility”, 27 of these were in Europe and Northern America.
- Maternity leave with job security was present in all 81 low-fertility countries (apart from the United States of America).
- 88% of 81 low-fertility countries, including Australia and Norway, had subsidised childcare.

^{dddd} Low fertility refers to a total fertility rate below 2.1 births. ¹⁹⁰

^{eeee} The goals of these types of family policies varies, including reducing income and gender inequalities and in some cases, promoting increases in the birthrate through “baby bonuses”.^{192,193}

^{ffff} Some academics suggest that supporting work and family life effectively, involves universally accessible public support throughout childhood.¹⁹³

^{gggg} Some academics argue that very few policies can substantially affect the birthrate.

^{hhhh} In 2022 the World Health Organisation noted that everyone has the right to:

- To choose whether to have children or not.
- To decide freely the number, spacing and timing of children.
- To get the information, education and support to make these decisions.¹⁹⁷

- 78% of low-fertility countries, including Hungary, provided child/family allowances.

The report also stated that, although some European countries have maintained fertility levels or seen slight increases, "there is a lack of evidence that these increases are stable and long term."¹⁹⁰

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